Hints: Linear algebra – 2. Vector spaces, dimension and rank

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. For the sum and the product, compare row spaces and column spaces: the rows of $A+B$ lie in row $A+$ row B, the columns of AB lie in $\operatorname{im} A$, and the rows of AB lie in the row space of B. For the block row, $\operatorname{im}(A \mid B) = \operatorname{im} A + \operatorname{im} B$.

Hint 2. Key lemma: restricting A to $\operatorname{im} X$ and applying rank–nullity gives $\operatorname{rank}(AX) = \operatorname{rank} X - \dim(\ker A \cap \operatorname{im} X)$.

Hint 3. Apply the lemma with $X = B$ and with $X = BC$, and use $\ker A \cap \operatorname{im}(BC) \subseteq \ker A \cap \operatorname{im} B$: this is Frobenius. Sylvester is Frobenius with middle matrix E_n. Alternatively, the lemma with $X = B$ gives $\operatorname{rank} AB - (\operatorname{rank} A + \operatorname{rank} B - n) = \dim \ker A - \dim(\ker A \cap \operatorname{im} B)$, which is 0 exactly when $\ker A \subseteq \operatorname{im} B$.

Hint 4. For sharpness use diagonal 0/1 matrices, i.e. coordinate projections E_S onto index sets $S \subseteq \{1, \dots, n\}$: $\operatorname{rank} E_S = |S|$ and $E_S E_T = E_{S \cap T}$. Sylvester: equality forces $a + b \geqslant n$, and then $S = \{1, \dots, a\}$, $T = \{n - b + 1, \dots, n\}$ give $|S \cap T| = a + b - n$. Frobenius with $A = E_S$, $B = E_{[b]}$, $C = E_T$ reads $|S \cap [b]| + |T \cap [b]| \leqslant b + |S \cap T \cap [b]|$, with equality iff $S \cup T \supseteq [b]$; this is possible with $|S| = a$, $|T| = c$ whenever $a + c \geqslant b$, and $a + c \geqslant b$ is necessary for equality anyway.

Problem 2

Hint 1. Collect the functionals into one linear map $F \colon V \to \mathbb{R}^n$, $F(x) = (f_1(x), \dots, f_n(x))$. The hypothesis says exactly $\ker F \subseteq \ker f$.

Hint 2. Hence $f(x)$ depends only on $F(x)$: $g(F(x)) = f(x)$ is a well-defined linear functional on the subspace $\operatorname{im} F \subseteq \mathbb{R}^n$. Extend it to all of $\mathbb{R}^n$ (extend a basis of $\operatorname{im} F$); every linear functional on $\mathbb{R}^n$ is $g(y) = c_1 y_1 + \dots + c_n y_n$. Then $f = g \circ F = c_1 f_1 + \dots + c_n f_n$.

Problem 3

Hint 1. Split a polynomial into homogeneous parts $P = \sum_k P_k$. The average of P_k over the circle of radius r is r^k times its average over the unit circle. So P is balanced if and only if every homogeneous part has zero average on the unit circle.

Hint 2. For odd k the average is automatically 0 (since $P_k(-x, -y) = -P_k(x, y)$). For even k, "average $= 0$" is one non-trivial linear condition, because $(x^2 + y^2)^{k/2}$ has average 1.

Hint 3. So $\dim V = \binom{2011}{2} - 1005$, subtracting one condition for each even $k \in \{0, \dots, 2008\}$. The answer is 2020050.

Problem 4

Hint 1. $A = u\mathbf{1}^T + \mathbf{1}u^T$ with $u = (1, \dots, n)^T$, so the rank is at most 2. Alternatively, subtract consecutive rows.

Hint 2. For $n \geqslant 2$ the two rows $(2, 3, \dots)$ and $(3, 4, \dots)$ are not proportional. The answer: rank 2 for $n \geqslant 2$, and 1 for $n = 1$.

Problem 5

Hint 1.

(a) Write A as a sum of r rank-one matrices $u_k v_k^T$ (e.g. take a basis of the column space). Put the u_k as the columns of B and the v_k^T as the rows of C. Then $\operatorname{rank} BC \leqslant \min(\operatorname{rank} B, \operatorname{rank} C) \leqslant r$ forces equality.

Hint 2.

(b) Let $D = CB \in M_r$. Then $A^{k+1} = BD^kC$. Take p the characteristic polynomial of D (degree r) and consider $q(x) = x\,p(x)$; Cayley–Hamilton gives $q(A) = Bp(D)C = 0$.

Problem 6

Hint 1. Put $S = AB+BA$ and use $B^2 = B$. Compute $(B-E)S$, $S(B-E)$ and $(B-E)S(B-E)$.

Hint 2. You get $(B-E)S = BAB - AB$, $S(B-E) = BAB - BA$, and $(B-E)S(B-E) = 0$. Apply Frobenius' inequality to the triple $B-E$, S, $B-E$, and note that $AB-BA$ is the difference of the first two matrices.

Problem 7

Hint 1. Show first that $f(PXQ) = f(X)$ for invertible P, Q: apply the inequality to PXQ and to $X = P^{-1}(PXQ)Q^{-1}$.

Hint 2. Hence $f(X) = g(\operatorname{rank} X)$ with $g(r) = f(\operatorname{diag}(E_r, 0))$. From $J_r J_s = J_r$ for $r \leqslant s$, g is non-decreasing, and it is surjective onto $\{0, \dots, n\}$. Hence $g = \mathrm{id}$: the only such function is $f(X) = \operatorname{rank} X$.

Problem 8

Hint 1. Consider $S = A_1 + \cdots + A_k$. What does anticommutation do to S^2?

Hint 2. $S^2 = \sum A_i^2 + \sum_{i<j}(A_iA_j + A_jA_i) = S$, so S is idempotent. For idempotents, rank equals trace, so $n \geqslant \operatorname{rank} S = \operatorname{tr} S = \sum \operatorname{rank} A_i$. Finish with pigeonhole.